

Generic Degree Three in $J\mathcal{C}(3)$: Collision Geometry and Galois Group S_3

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Abstract

For a polynomial self-map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n], \quad I_R(F) = (F_i(x) - F_i(y))_{i=1}^n.$$

The quotient $C_F = S/I_R(F)$ is the coordinate ring of the collision scheme of pairs with the same image. The image in C_F of the diagonal ideal $I_{\Delta} = (x_i - y_i)_{i=1}^n$ is the obstruction ideal $\text{Obs}(F)$, canonically identified with $I_{\Delta}/I_R(F)$. Equivalently, $\text{Obs}(F) = \ker(\bar{\mu}_F)$, where $\bar{\mu}_F: C_F \rightarrow \mathbb{C}[x_1, \dots, x_n]$ is diagonal evaluation. Over $\mathbb{C}$, vanishing of this ideal characterizes polynomial automorphisms.

We analyze this obstruction in generic degree three. Put $B = \mathbb{C}[F_1, \dots, F_n]$, $K = \text{Frac}(B)$, and $L = \mathbb{C}(x_1, \dots, x_n)$. Generic base change identifies the generic collision algebra with $L \otimes_K L$. For a chosen primitive generator α , the marked-root decomposition gives a quadratic residual L -algebra D_{α} and equivalences

$$K \otimes_B C_F \simeq_K L \otimes_K L, \quad L \otimes_K L \simeq_L L \times D_{\alpha},$$

with tensor multiplication equal to first projection. This decomposition applies to any finite separable generic extension and requires no Keller hypothesis. The Keller condition, though unnecessary for the decomposition, makes F étale and generically finite, so every possible counterexample lies in this finite separable setting. Classical Galois rigidity excludes generic degrees one and two; generic degree three is therefore the first possible counterexample degree. Specialize now to $n = 3$. Since $[L : K] = 3$, the map F cannot be a polynomial automorphism. Thus, on the generic-degree-three locus,

$$F \text{ is a counterexample to } J\mathcal{C}(3) \iff F \text{ is Keller.}$$

The residual algebra is split in the normal cubic branch and connected in the nonnormal branch. Classical Galois rigidity excludes the normal branch from the counterexample locus. Consequently, if this locus is nonempty, then, with N denoting the normal closure of L/K ,

$$D_{\alpha} \simeq_L N, \quad K \otimes_B C_F \simeq L \otimes_K L \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

Thus the residual generic collision factor of every such counterexample is the normal closure N , whose Galois group is S_3 . No existence assertion is made.

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1 Introduction

Let $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ be a polynomial self-map. Write

$$R_F := \mathbb{A}_{\mathbb{C}}^n \times_{\mathbb{A}_{\mathbb{C}}^n} \mathbb{A}_{\mathbb{C}}^n,$$

where both maps to the middle copy are F . This is the collision scheme: its points are ordered pairs with the same image under F , and the diagonal defines a canonical closed subscheme of R_F . We ask how the remaining generic factors are organized around it.

Our end goal is to describe the generic-degree-three counterexample locus for the complex three-dimensional Jacobian conjecture. Degree three already rules out polynomial automorphy, so within this locus the counterexample condition is equivalent to the Keller condition. Classical Galois rigidity then eliminates the normal cubic branch. Every possible counterexample in this degree class is therefore nonnormal; its quadratic residual factor is the normal closure with Galois group S_3 . This is the conclusion toward which the paper is organized.

For a polynomial map F , the collision and diagonal ideals satisfy $I_R \subseteq I_{\Delta}$. The quotient

$$\text{Obs}(F) = I_{\Delta}/I_R \subseteq C_F = S/I_R$$

is the kernel of diagonal evaluation. Consequently,

$$\text{Obs}(F) = 0 \iff I_R = I_{\Delta} \iff F \text{ is a polynomial automorphism.}$$

This gives a single affine ideal whose support is the failure of the collision relation to collapse to the diagonal.

The generic fiber makes the field-theoretic content transparent. Put

$$B = \mathbb{C}[F_1, \dots, F_n], \quad K = \text{Frac}(B), \quad L = \mathbb{C}(x_1, \dots, x_n).$$

Under the usual dominance and generic-finiteness hypotheses, base change to K identifies the collision algebra with

$$K \otimes_B C_F \simeq L \otimes_K L,$$

and diagonal evaluation with tensor multiplication. A primitive element of a finite separable extension then exhibits the marked root as the first factor and the residual conjugate algebra as

the second. These finiteness and separability hypotheses are automatic for Keller maps. Galois rigidity then excludes degrees one and two for a counterexample, making generic degree three the first case to analyze.

Suppose now that $[L : K] = 3$. The marked-root decomposition splits $L \otimes_K L$ into its diagonal factor and a quadratic residual L -algebra D_α . For a general cubic map, this algebra is split in the normal branch and connected in the nonnormal branch. The degree assumption already makes F nonautomorphic, so F is a counterexample to $JC(3)$ if and only if it is Keller. On this counterexample locus, classical Galois rigidity excludes the split branch. Thus D_α is connected, it is the normal closure N , and

$$L \otimes_K L \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

Under the chosen power-basis presentation, the canonical diagonal factor corresponds to α , while the residual factor supplies a second root; in a cubic extension those two roots determine the third. The faithful action of $G = \text{Gal}(N/K)$ on the three cosets of $H = \text{Gal}(N/L)$ gives the permutation representation; the degree count identifies this Galois group with S_3 . The nonzero residual factor is the collision-theoretic witness of nonautomorphy. We do not assert that the generic-degree-three counterexample locus is nonempty, or that every possible counterexample has generic degree three.

Section 2 develops the collision ideal, fiber-product interpretation, generic base change, and marked-root decomposition. Section 3 identifies the two cubic residual branches and shows that the counterexample locus can occur only in the nonnormal branch, whose normal closure has Galois group S_3 . Section 4 records the dimension-independent normalization picture that motivates a planar sequel.

2 General construction

The organizing question is how the failure of a polynomial self-map to be one-to-one is encoded by the part of its fiber square lying off the diagonal. The collision ideal isolates that part scheme-theoretically; generic base change then identifies its field-level factors with conjugate embeddings. All constructions in this section are dimension-independent.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n): X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^\times.$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \quad I_\Delta = (x_i - y_i)_{i=1}^n, \quad \text{Obs}(F) := I_\Delta / I_R(F).$$

The quotient $\text{Obs}(F)$ is naturally an ideal of the collision ring $C_F := S / I_R(F)$; we call it the obstruction ideal. For a fixed map F , we abbreviate $I_R(F)$ to I_R .

Proposition 2.1 (Canonical containment). *For every polynomial map F ,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H , or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i . $\square$

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F: C_F \longrightarrow A, \quad [s]_{I_R} \longmapsto s(x, x).$$

Proposition 2.2 (The diagonal kernel). *As ideals of C_F ,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The kernel of diagonal evaluation $S \rightarrow A$ is I_Δ . Passing through S/I_R therefore identifies the displayed kernel with I_Δ/I_R . Since $I_R \subseteq I_\Delta$,

$$I_\Delta/I_R = 0 \iff I_R = I_\Delta.$$

$\square$

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \text{Spec } A$, let $Y = \text{Spec } B$, and let

$$f_F: X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y: A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F . Their common restriction equips C_F with a B -algebra structure.

Theorem 2.3 (Fiber-product interpretation). *There is a canonical B -algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \quad R_F := \text{Spec } C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \quad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \rightarrow T$ to a commutative $\mathbb{C}$ -algebra T is equivalent to a pair of maps $f, g: A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \quad (1 \leq i \leq n).$$

Since the F_i generate B , this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\text{Spec}(A \otimes_B A) \cong \text{Spec } A \times_{\text{Spec } B} \text{Spec } A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. $\square$

On complex points,

$$R_F(\mathbb{C}) = \{(u, v) : F(u) = F(v)\}.$$

Inside $X \times X = \text{Spec } S$, the ideals I_R and I_Δ define $X \times_Y X$ and its diagonal. Thus

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes. This is the geometric form of obstruction vanishing.

2.3 Generic fibers and marked-root decomposition

Let F be a polynomial self-map, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Proposition 2.4 (Keller generic geometry). *If F is Keller, then its coordinate functions are algebraically independent, the morphism*

$$f_F: \text{Spec } A \longrightarrow \text{Spec } B$$

is étale and quasi-finite, and L/K is a finite separable extension. Consequently, every Keller map, and in particular every complex Jacobian-conjecture counterexample, has a finite separable generic field extension.

Proof. The identity

$$dF_1 \wedge \dots \wedge dF_n = \det(JF) dx_1 \wedge \dots \wedge dx_n \neq 0$$

gives algebraic independence, so $B \cong \mathbb{C}[u_1, \dots, u_n]$. As a B -algebra, A has the presentation

$$A \cong \mathbb{C}[u_1, \dots, u_n, X_1, \dots, X_n] / (F_i(X) - u_i)_{i=1}^n.$$

Its relative Jacobian determinant is the unit $\det(JF)$. The Jacobian criterion therefore makes f_F étale, hence quasi-finite. Since $B \subseteq A$, the morphism is dominant, so quasi-finiteness gives $[L : K] < \infty$. Its generic point is étale, and therefore L/K is separable. $\square$

For the remainder of this subsection, assume that F is dominant and generically finite. Let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 2.5 (Generic collision bridge). *The map θ is an isomorphism. Consequently, there is a K -algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \quad \frac{b}{u} \otimes a \mapsto \frac{ba}{u}.$$

Since $A = \mathbb{C}[x_1, \dots, x_n]$ is a domain and $B \subseteq A$,

$$B \setminus \{0\} \subseteq A \setminus \{0\}.$$

Hence

$$U^{-1}A \hookrightarrow \text{Frac}(A) = L, \quad \frac{a}{u} \mapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives θ , so θ is injective.

Since F is generically finite, L/K is a finite extension. The image of $U^{-1}A$ in L is

$$K[x_1, \dots, x_n].$$

Since each x_i is algebraic over K , this K -domain is finite-dimensional over K , hence is a field. Since it contains A , it contains $\text{Frac}(A) = L$; the reverse inclusion is already given by its embedding into L . Therefore

$$U^{-1}A = L,$$

and θ is an isomorphism.

By Theorem 2.3, identify C_F with $A \otimes_B A$. There is a K -algebra isomorphism

$$\beta: K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \mapsto \lambda \lambda' \otimes (a \otimes a').$$

The balancing relations over B and K show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing β with $\theta \otimes_K \theta$ gives

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under $C_F \cong A \otimes_B A$, the homomorphism $\bar{\mu}_F$ is multiplication. Therefore, on a pure tensor,

$$\begin{aligned} (\mu_L \circ \Phi_F)(\lambda \otimes (a \otimes a')) &= (\lambda a) a' = \lambda a a', \\ (\theta \circ (1 \otimes \bar{\mu}_F))(\lambda \otimes (a \otimes a')) &= \theta(\lambda \otimes a a') = \lambda a a'. \end{aligned}$$

Pure tensors generate $K \otimes_B C_F$, so the two homomorphisms are equal. This proves the commutative square. $\square$

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α , and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Theorem 2.6 (Marked-root tensor decomposition). *Give $L \otimes_K L$ its L -algebra structure through the left tensor factor. There is an L -algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \rightarrow L$ is projection onto the first factor.

Proof. The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from K to L give an L -algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \quad 1 \otimes \alpha \mapsto T.$$

Since m_α is separable, $m'_\alpha(\alpha) \neq 0$. Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at $T = \alpha$ gives

$$h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0.$$

Thus $T - \alpha$ does not divide h_α , so these two polynomials are coprime in $L[T]$. The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor $\ell \otimes p(\alpha)$, with $p \in K[T]$, corresponds to the class of $\ell p(T)$. Tensor multiplication sends it to $\ell p(\alpha)$, which is evaluation at $T = \alpha$. Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor $L[T]/(T - \alpha) \cong L$. Hence tensor multiplication is the first projection. $\square$

Together with Theorem 2.5, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

2.4 Automorphism detection

Theorem 2.7 (Automorphism criterion). *For an arbitrary polynomial self-map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, the following are equivalent:*

- (1) *F is a polynomial automorphism;*
- (2) *F is injective on $\mathbb{A}^n(\mathbb{C})$;*
- (3) *$I_R(F) = I_\Delta$;*

(4) $\text{Obs}(F) = 0$.

Proof. Suppose first that F has polynomial inverse $G = (G_1, \dots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i , so $I_\Delta \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_\Delta$, and let $\text{ev}_{(a,b)}: S \rightarrow \mathbb{C}$ be evaluation at (a, b) . If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\text{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_\Delta = I_R$ gives $a_i - b_i = 0$ for every i . Thus $a = b$, proving (3) $\Rightarrow$ (2). The Ax–Grothendieck theorem [Ax69] gives (2) $\Rightarrow$ (1). Finally, Proposition 2.2 gives (3) $\iff$ (4). $\square$

Applied to Keller maps, Theorem 2.7 says that $JC(n)$ is equivalent to universal vanishing of $\text{Obs}(F)$, or equivalently to $I_R(F) = I_\Delta$.

We will also use the classical Galois case of the conjecture. It is the step that rules out the normal cubic branch under the Keller hypothesis.

Theorem 2.8 (Classical Galois rigidity). *Let*

$$F: \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^\times,$$

and put

$$K = \mathbb{C}(F_1, \dots, F_n), \quad L = \mathbb{C}(x_1, \dots, x_n).$$

If L/K is normal, then F is a polynomial automorphism and $L = K$.

Proof. The Keller condition makes L/K finite and separable. Thus normality makes it Galois, and the conclusion is the classical Galois case of the Jacobian conjecture due to Campbell, with algebraic treatments by Razar and Wright [Cam73, Raz79, Wri81]. $\square$

Lean correspondence. `CollisionIdeals/General/Collision/CollisionDiagonal.lean`,
`CollisionIdeals/General/FiberProduct/Polynomial.lean`,
`CollisionIdeals/General/GenericFiber/CollisionDiagonal.lean`,
`CollisionIdeals/General/GenericFiber/MarkedRootDecomposition.lean`,
`CollisionIdeals/General/Keller/Interfaces.lean`, and
`CollisionIdeals/General/Automorphism/Equivalences.lean`. The Lean form of Theorem 2.5 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness. The predicates `KellerEtaleBridge` and `KellerFlatBridge` expose parts of Proposition 2.4 as theorem targets; Lean does not yet derive them, generic finiteness, or separability from `IsKeller`.

3 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to generic degree three. After the diagonal is exhibited as the first factor, the complementary generic collision algebra has degree two over L . For an arbitrary cubic map it is either split, corresponding to a normal cubic extension, or connected, corresponding to a nonnormal cubic extension. Degree three already rules out polynomial automorphy, so within this locus the counterexamples to $JC(3)$ are precisely the Keller maps. Classical Galois rigidity rules out the split branch on that counterexample locus. Its residual factor is therefore the normal closure, whose Galois group is S_3 .

Let $S_3 = \text{Perm}(\{1, 2, 3\})$ be the symmetric group on three letters.

3.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \quad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 2.5. Assume that F has generic degree three:

$$[L : K] = 3.$$

Because K has characteristic zero, L/K is separable. Choose a primitive element α [The26], so that $L = K(\alpha)$ with power basis $1, \alpha, \alpha^2$. If $m_\alpha \in K[T]$ is the minimal polynomial of α , let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

The marked-root decomposition identifies the algebra remaining after the diagonal factor with

$$D_\alpha := L[T]/(h_\alpha),$$

a finite L -algebra of degree two. Let N/K be the normal closure of L/K , containing L through the marked inclusion. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Restriction of automorphisms gives a bijection

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \longmapsto g|_L.$$

Because the conjugates of L generate its normal closure N , an element of G fixing every conjugate embedding fixes N . Hence the coset action is faithful and

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

The residual collision algebra detects which cubic branch occurs. We include the short argument in order to fix its marked identification with the normal closure.

Proposition 3.1 (Connected cubic residual factor). *The following conditions are equivalent:*

- (i) $\text{Spec } D_\alpha$ is connected;
- (ii) D_α is a field;
- (iii) h_α is irreducible over L ;
- (iv) L/K is nonnormal.

When they hold, there is an L -algebra isomorphism

$$\varphi: D_\alpha \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

The quadratic h_α is separable. If it is irreducible, then D_α is a field and hence has connected spectrum. If it is reducible, it is a product of two distinct linear factors; the Chinese remainder theorem then gives $D_\alpha \simeq L \times L$, whose spectrum is disconnected. This proves the equivalence of (i), (ii), and (iii).

If h_α is reducible, it has a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root is

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L . Since $L = K(\alpha)$, the extension L/K would then be the splitting field of the separable polynomial m_α , hence normal. Conversely, if L/K is normal, then m_α splits over L , so h_α is reducible. Hence (iii) and (iv) are equivalent.

Assume these equivalent conditions, and let β denote the image of T in D_α . This field contains the three roots

$$\alpha, \quad \beta, \quad -a_2 - \alpha - \beta$$

of m_α , and is generated over L by β . It is therefore the splitting field of m_α , which is the normal closure N of L/K . Choose the resulting L -algebra isomorphism

$$\varphi: D_\alpha \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha: L \otimes_K L \xrightarrow{\sim}_L L \times D_\alpha$$

be the marked-root decomposition of Theorem 2.6. It satisfies

$$\text{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with φ , let

$$\Psi = (\text{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since $\text{id}_L \times \varphi$ leaves the first factor unchanged, $\text{pr}_1 \circ \Psi = \mu_L$. □

Corollary 3.2 (The generic-degree-three counterexample locus). *The map F is not a polynomial automorphism. Consequently,*

$$F \text{ is a counterexample to } JC(3) \iff F \text{ is Keller.}$$

When these equivalent conditions hold, L/K is nonnormal and $\text{Spec } D_\alpha$ is connected.

Proof. If F were a polynomial automorphism, its coordinate functions would generate A , so $B = A$ and $L = K$, contrary to $[L : K] = 3$. The displayed equivalence now follows from the definition of a counterexample to $JC(3)$.

Assume these conditions, so that F is Keller. If L/K were normal, Theorem 2.8 would give $L = K$, again contradicting $[L : K] = 3$. Proposition 3.1 now identifies nonnormality with connectedness of $\text{Spec } D_\alpha$. $\square$

Theorem 3.3 (Galois group of the connected residual factor). *If $\text{Spec } D_\alpha$ is connected, then*

$$\text{Gal}(N/K) \cong S_3.$$

Proof. Proposition 3.1 identifies N with the degree-two field D_α over L . Hence

$$|H| = [N : L] = 2.$$

The normal-closure identities recalled above give

$$\text{core}_G(H) = 1, \quad [G : H] = [L : K] = 3.$$

Hence the action of G on the three cosets of H gives

$$\rho: G \longrightarrow \text{Perm}(G/H) \cong S_3, \quad \ker \rho = \text{core}_G(H) = 1.$$

Core-freeness therefore makes ρ injective, so

$$|G| \leq |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. $\square$

3.2 The cubic residual factor

Retain the cubic data of Section 3.1 and suppose that F lies in the counterexample locus of Corollary 3.2. Equivalently, F is Keller; the same corollary shows that D_α is connected and L/K is nonnormal. Proposition 3.1 then identifies D_α with N . Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \quad K = \text{Frac}(B) \subseteq L = \text{Frac}(A) = K(\alpha) \subseteq N,$$

where $[L : K] = 3$ and N/K is the normal closure. Let

$$\theta: K \otimes_B A \xrightarrow{\sim} L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Corollary 3.4 (Generic cubic S_3 collision). *There is a K -algebra isomorphism*

$$\Psi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \quad \text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \quad \text{Gal}(N/K) \cong S_3.$$

If

$$C_{F,K} := K \otimes_B C_F, \quad J_{F,K} := \ker(\theta \circ (1 \otimes \bar{\mu}_F)),$$

then the intrinsic residual quotient of the generic collision algebra is

$$C_{F,K} / \text{Ann}_{C_{F,K}}(J_{F,K}) \simeq_K N.$$

Consequently,

$$\text{Obs}(F) \neq 0, \quad I_R \subsetneq I_\Delta, \quad F \notin \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Proof. Theorem 2.5 gives a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 3.1 gives

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K , let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \quad \Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N \neq 0.$$

The annihilator of $0 \times N$ in $L \times N$ is $L \times 0$. Therefore Ψ_F , followed by the second projection, induces

$$C_{F,K} / \text{Ann}_{C_{F,K}}(J_{F,K}) \xrightarrow{\sim}_K N.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F: C_F \rightarrow A$ were injective. Since K is a localization of B , it is flat over B , so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 2.5 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 2.2 gives $\text{Obs}(F) \neq 0$. Proposition 2.1 shows that $I_R \subseteq I_\Delta$ is strict.

Finally, Theorem 3.3 identifies $\text{Gal}(N/K)$ with S_3 . If F were a polynomial automorphism, Theorem 2.7 would give $\text{Obs}(F) = 0$, contrary to $\text{Obs}(F) \neq 0$. $\square$

Remark (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2 T^2 + a_1 T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N . For distinct i, j, k ,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \quad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the residual factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/Cubic/GaloisGroup.lean`,
`CollisionIdeals/ComplexThree/Cubic/S3Collision.lean`,
`CollisionIdeals/ComplexThree/Cubic/Main.lean`, and
`CollisionIdeals/ComplexThree/Statements/JacobianConjecture.lean`. The generic product-annihilator quotient is formalized in `CollisionIdeals/General/GenericFiber/DiagonalObstruction.lean`. The Lean cubic witness takes the residual-field isomorphism and $H \neq 1$ as explicit inputs. It proves obstruction nonvanishing before using the Keller hypothesis solely to assemble the counterexample statement; the exact generic-degree-three equivalence displayed above is not yet packaged as a separate Lean theorem. In the manuscript the two field-theoretic inputs follow from Proposition 3.1 and Corollary 3.2; the use of classical Galois rigidity in that corollary is not yet formalized. Hidden inertia is formalized separately in `CollisionIdeals/ComplexThree/Cubic/HiddenInertia.lean`, so the cubic core has no normalization or ramification dependency.

Remark (Exact relation with classical complex $JC(3)$). The traditional complex three-dimensional Jacobian conjecture asserts that every polynomial map

$$F: \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3, \quad \det JF \in \mathbb{C}^\times,$$

is a polynomial automorphism; equivalently, every such map satisfies $\text{Obs}(F) = 0$. In generic degree three, Corollary 3.2 instead gives

$$\begin{aligned} [L : K] = 3 &\implies \text{Obs}(F) \neq 0, & F \notin \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3), \\ F \text{ is a counterexample to } JC(3) &\iff \det JF \in \mathbb{C}^\times, \\ F \text{ is a counterexample} &\implies \text{Spec } D_\alpha \text{ connected,} & \text{Gal}(N/K) \cong S_3. \end{aligned}$$

The first line is independent of the Keller condition. The additional content on the counterexample locus is that Theorem 2.8 excludes the normal cubic branch, after which the residual algebra itself produces the normal closure and its Galois group S_3 . Thus $JC(3)$ implies that the generic-degree-three counterexample locus is empty. This section does not construct such a map, and it does not assert that every possible counterexample to $JC(3)$ has generic degree three.

4 Further directions

The normal closure also carries a dimension-independent boundary picture, although none of it is needed for the cubic collision theorem. Let L/K be finite separable, let N/K be its normal closure, and let Y be a normal integral model with function field K . Write

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L), \quad Z = \text{Norm}_N(Y) \longrightarrow \overline{X} = \text{Norm}_L(Y) \longrightarrow Y.$$

Suppose that a normal model X of L is an open subscheme of $\overline{X}$ over Y , and that $X \rightarrow Y$ is étale.

Proposition 4.1 (Hidden inertia on a conjugate sheet). *Let $E \in Z^{(1)}$ be ramified over Y , with divisorial valuation w_E , restriction $v = w_E|_K$, and inertia group I_E . Then some $g \in G$ satisfies*

$$e((g^{-1}w_E)|_{L/v}) = [I_E : I_E \cap gHg^{-1}] > 1.$$

For every such g , the center of $(g^{-1}w_E)|_L$ on $\overline{X}$ lies in $\overline{X} \setminus X$.

Proof. Because N is the normal closure of L/K , the action of G on G/H is faithful, so $\bigcap_{g \in G} gHg^{-1} = 1$. Since $I_E \neq 1$, it is not contained in every conjugate of H ; the usual inertia formula gives the displayed index for a suitable g . If the corresponding center belonged to X , the étale morphism $X \rightarrow Y$ would force its ramification index to be one, a contradiction. $\square$

On the generic-degree-three counterexample locus of Section 3, the cubic extension is necessarily nonnormal and $G \cong S_3$. Thus any divisorial ramification in the ordered-root normalization is detected on a conjugate sheet and forced into the deleted boundary. This statement is conditional on the existence of such a ramified divisor; the cubic collision theorem itself neither requires nor asserts it.

The cubic and planar problems use the same diagonal in opposite ways. Here generic degree three forces a nonzero residual factor beside it. In the planar sequel, the goal is to prove that the off-diagonal factor vanishes, so that the collision scheme consists only of the diagonal.

A sequel will develop this normalization picture for planar Keller maps: conjugate affine sheets, their boundary and fixed-locus ideals, and the secant–trace comparison. Its remaining Keller-specific input is a uniform landing theorem converting a finite secant–frame lattice into control of arbitrary boundary sections. Purity and finite-étale rigidity enter only after that boundary statement is established.

5 Conclusion

The collision ideal turns injectivity into a scheme-theoretic equality. The obstruction $\text{Obs}(F)$ is precisely the kernel of diagonal evaluation, and it vanishes exactly when the collision relation equals the diagonal. After passage to the generic fiber, that diagonal becomes the multiplication factor of $L \otimes_K L$, while the remaining factors record the other embeddings of L .

In generic degree three, the diagonal persists as the first factor beside a quadratic residual algebra, and nonautomorphy is automatic. Thus the counterexample condition is equivalent to the Keller condition. On this locus the classical Galois case rules out the split, normal branch; hence the residual algebra is connected, it is the normal closure N , its Galois group is S_3 , and

$$K \otimes_B C_F \simeq L \times N.$$

Its nonzero residual factor gives $\text{Obs}(F) \neq 0$. Hence every map in the generic-degree-three counterexample locus has the normal-closure factor N , with $\text{Gal}(N/K) \cong S_3$. The theorem does not assert that this locus is nonempty or classify counterexamples of other generic degrees. The normalization principle of Section 4 explains how the same conjugate sheet geometry leads to boundary questions, which are reserved for the planar sequel.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result describes the structure of the generic-degree-three counterexample locus.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

Corollary 5.1 (First possible generic degree). *If F is Keller and $[L : K] \leq 2$, then F is a polynomial automorphism. Consequently, every complex Jacobian-conjecture counterexample has generic degree at least three.*

Proof. Proposition 2.4 makes L/K finite and separable. In degree one it is trivial, and in degree two it is Galois. Theorem 2.8 applies in either case. $\square$

This low-degree boundary is the handoff to the planar sequel. Paper II asks for the stronger conclusion that every planar Keller map has generic degree one—equivalently, by Galois rigidity and the automorphism criterion, that its collision scheme collapses to the diagonal.

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